Precipitation Retrieval Integrating Multiple Satellite Observations: A Dataset and a Framework

Zheng Wang, Boxian He, Chunjiao Wang, Bin Xu, and Cong Bai, *Member, IEEE*

***Abstract*— Multimodal satellite observations have been widely used for precipitation retrieval. Numerous retrieval algorithms and precipitation products have been developed based on these data. However, the integrated retrieval of multimodal data remains challenging due to the modality heterogeneity caused by different data characteristics reflecting precipitation patterns. To address these issues, effectively integrating multimodal data is crucial. We propose a framework named precipitation retrieval integrating multiple satellite observations and geographical information (PRMG), consisting of two networks for precipitation identification and estimation, respectively. Specifically, PRMG includes a multibranch fusion (MBF) module for integrating three types of satellite observations: infrared (IR), passive microwave (PMW), and spaceborne precipitation radar (PR), and a geographical information correction (GIC) module to incorporate the geographical information to calibrate precipitation features. We also collect a new precipitation retrieval dataset for multimodal precipitation retrieval, called Precipitation-MG, which includes satellite observations, corresponding geographical information, and precipitation products. Extensive experiments on Precipitation-MG demonstrate the effectiveness of the multimodal fusion method and the geographic correction method. The retrieval performance of PRMG achieves significant improvements compared to the Global Precipitation Measurement (GPM) currently in operation, i.e., Level-2 dual-frequency precipitation radar (DPR) and GPM microwave imager (GMI) combined (2B-CMB) product. The source code and dataset are publicly available at https://github.com/Zjut-MultimediaPlus/PRMG**

***Index Terms*— Deep learning, geographic correction, multimodal data fusion, precipitation retrieval.**

I. Introduction

PRECIPITATION is a critical part of the Earth's water cycle, and its accurate measurement and estimation are of great importance to meteorological and hydrological research, including weather forecasting, flood and drought warnings, and long-term climate change studies [1]. Traditional precipitation observations rely on rain gauges and ground-based radars. However, the deployment density and observational range of these instruments are often constrained by challenging geographical conditions. For instance, in remote and topographically complex regions, the deployment of rain gauges and ground-based radars is extremely sparse, making it difficult for ground-based observations to accurately capture widespread precipitation distributions.

Satellite observations offer broad coverage and can overcome the limitations of ground-based observations. Three main types of satellite observations are commonly used for precipitation retrieval: infrared (IR), passive microwave (PMW), and spaceborne precipitation radar (PR) reflectivity, as shown in Fig. 1. IR data from geostationary Earth orbit (GEO) satellites provide quasi-global coverage and high spatiotemporal resolution. They capture the relationship between bright,

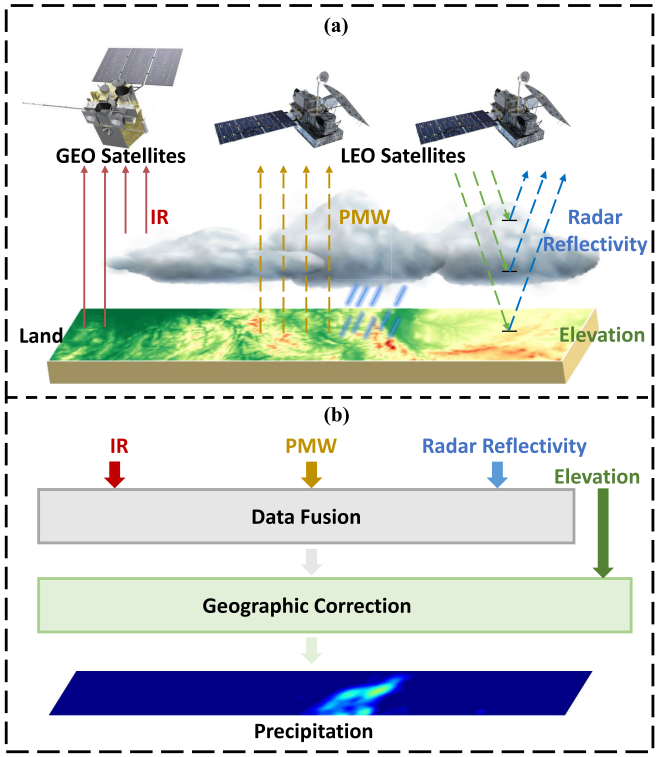


Fig. 1. Illustrations of (a) multimodal satellite observations and their working principle and (b) multimodal data of IR, PMW, radar reflectivity, and elevation are fused to achieve highly accurate precipitation retrieval.

Received 23 September 2024; revised 21 April 2025; accepted 7 June 2025. Date of publication 3 July 2025; date of current version 15 July 2025. This work was supported in part by Zhejiang Provincial Natural Science Foundation of China under Grant LRG25F020002; in part by the Project of China Meteorological Service Association under Grant CMSA2023MD001; in part by the Satellite Application Advance Plan of Feng-Yun under Grant FYAPP-2022.0608; in part by the Natural Science Foundation of China under Grant 62302453; and in part by China Meteorological Administration Key Innovation Team under Grant CMA2023ZD01. *(Corresponding author: Cong Bai.)*

Zheng Wang and Boxian He are with the College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou 310023, China (e-mail: zhengwang@zjut.edu.cn; 221122120152@zjut.edu.cn).

Chunjiao Wang and Bin Xu are with the National Meteorological Information Center, Beijing 100081, China (e-mail: wangcj@cma.gov.cn; xubin@cma.gov.cn).

Cong Bai is with the College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou 310023, China, and also with Zhejiang Key Laboratory of Visual Information Intelligent Processing, Hangzhou 310023, China (e-mail: congbai@zjut.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3585407

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cold clouds, and convection, using cloud-top temperature to estimate precipitation [2]. However, IR-based algorithms are limited by their inability to detect precipitation-related information within and below clouds. PMW data from low Earth orbit (LEO) satellites offer better physical properties than IR data, as they can penetrate clouds and capture particles that may produce precipitation [3]. Although PMW provides more direct measurements of precipitation, its spatiotemporal resolution is much lower than that of IR. Spaceborne PR reflectivity data from PRs carried by a few LEO satellites, on the other hand, can capture the 3-D microphysical structure of precipitation, providing high-quality cloud and precipitation characteristics [4], [5]. However, PR reflectivity observations are limited to relatively narrow swaths and are subject to issues like beam blockage and anomalous propagation errors [6].

Integrating multimodal data for precipitation retrieval (i.e., integrated retrieval) or fusing multisource precipitation rate can leverage the complementary strengths of each data. The Tropical Rainfall Measuring Mission (TRMM) and Global Precipitation Measurement (GPM) Mission have conducted extensive research in this field. Traditional algorithms, such as the Climate Prediction Center (CPC) morphing technique (CMORPH) algorithm [7], the TRMM Multisatellite Precipitation Analysis (TMPA) algorithm [8], and integrated multisatellite retrievals for GPM (IMERG) algorithm [9], use multimodal data or fusing multisource precipitation rate to produce satellite precipitation products (SPPs). However, these traditional algorithms rely on statistical or physical methods, requiring numerous variables, high computational power, and significant data storage. Many SPPs exhibit systemic biases and uncertainties related to modality heterogeneity caused by differences in satellite sensors, spatiotemporal resolution, and data characteristics [10], [11], [12].

In recent years, deep learning has increasingly been applied to precipitation retrieval [13], [14], [15]. Compared to traditional methods, deep learning approaches for integrated retrieval require fewer computational and storage resources and are more flexible in data fusion, enabling more effective use of multimodal data for precise precipitation retrieval. However, most methods focus on using deep learning to intercalibrate multiple SPPs or correcting the bias of SPPs with satellite observations, which can introduce inherent system biases and risk error accumulation. Some methods establish end-to-end models between multimodal satellite observations and precipitation, but they often simply concatenate observations, leading to insufficient data fusion. In this work, we utilize raw satellite observations instead of SPPs and design an effective multimodal fusion method to integrate the data, extracting complementary information and addressing modality heterogeneity.

In addition, satellite precipitation retrieval is susceptible to the complex terrain of different regions [16], [17], [18]. SPPs often show errors in complex terrains (e.g., the western coastal mountains of the United States and the Tibetan Plateau of China) [19], [20]. Precipitation retrieval from multimodal satellite observations in such regions remains a key challenge in the GPM era. To account for the terrain's impact on precipitation, many deep learning methods incorporate geographical information as an additional input or constraint, primarily to correct SPP biases. However, simple interactions between geographical information and SPP-derived precipitation rates pose limitations for bias correction [21]. Few methods integrate geographical information with satellite observations for feature construction. Therefore, we integrate geographical information with satellite observations at the feature level to correct representations of terrain-influenced precipitation patterns.

In this work, we propose a precipitation retrieval framework called precipitation retrieval integrating multiple satellite observations and geographical information (PRMG). PRMG consists of two networks dedicated to precipitation identification and estimation, respectively. Specifically, we design a multibranch fusion (MBF) module to fuse three types of satellite observations and a geographical information correction (GIC) module to address the impact of terrain on precipitation. We also construct a large-scale multimodal precipitation retrieval dataset called Precipitation-MG, encompassing comprehensive satellite and geographical data. Our contributions are summarized as follows.

1) PRMG framework is proposed, which effectively integrates three types of satellite observations and appropriately corrects the impact of terrain on precipitation. To the authors' knowledge, this is the first method to integrate these three types of satellite observations (i.e., IR, PMW, and PR) for precipitation retrieval.
2) A new large-scale multimodal precipitation retrieval database called Precipitation-MG is constructed, comprising 3886 LEO data records collected over a two-year period. Each record involves three types of satellite observations: IR, PMW, and PR reflectivity, three types of geographical information: latitude, longitude, and elevation, and along with two types of precipitation products: China Meteorological Administration Land Data Assimilation System (CLDAS-V2.0) and GPM 2B-CMB.
3) Extensive experiments are conducted to explore the optimal multibranch structure for integrating these observations. Additionally, we perform ablation studies to evaluate the contributions of the three modalities and the two proposed modules in precipitation retrieval. Finally, we demonstrate the superior performance of PRMG compared to the currently operational GPM 2B-CMB product.

II. Related Work

A. Traditional Methods of Precipitation Retrieval

Many statistical or physical precipitation retrieval algorithms based on satellite observations have been proposed. For IR data, Arkin and Meisner [22] proposed the geostationary operational environmental satellite (GOES) precipitation index (GPI) algorithm to establish an equation relating cloud-top temperature to precipitation. Ba and Gruber [23] proposed the GOES multispectral rainfall algorithm (GMSRA) algorithm based on GPI, which incorporates factors of humidity and cloud growth rate. For PMW data, Wilheit et al. [24] proposed a PMW-based algorithm first to establish a simple regression relationship between PMW brightness temperatures (BTs) and

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precipitation. Kummerow et al. [25] introduced the Goddard profiling (GPROF) algorithm, which uses Bayesian methods to match PMW BTs with precipitation profiles in an established cloud-radiation database for precipitation retrieval. For spaceborne PR reflectivity data, the PR-based algorithm of GPM uses Ku-band (13.6 GHz) and Ka-band (35.5 GHz) radar echoes to calculate the radar reflectivity factors and establish physical models related to raindrop size distribution (DSD) [26]. Arulraj and Barros [27] designed a vertical correlation structure (VCS) function to describe the structure of radar reflectivity profiles and demonstrated the potential for precipitation detection and intensity classification using multifrequency data from spaceborne PR.

Integrating multimodal data for precipitation retrieval or fusing multisource precipitation rate can take the advantage of complementary information of each data. CMORPH algorithm propagated and interpolated PMW-based precipitation estimates through cloud system advection vectors (CSAVs) computed from IR observation [7]. CMORPH-KF improved CMORPH based on Kalman filtering [28]. TMPA algorithm initially merges and calibrates multiple PMW-based precipitation estimations as ground truth (GT), and then builds up an empirical relationship between IR data and GT [8]. The IMERG algorithm inherits from TMPA and uses algorithms such as GPROF and CMORPH-KF to intercalibrate, merge, and interpolate IR, PMW, and PR precipitation estimates and ground-based precipitation measurements to obtain a unified globally gridded precipitation [9].

B. Deep Learning Methods of Precipitation Retrieval

Deep learning has been increasingly used in the field of precipitation retrieval. For IR data, Hong et al. [29] designed a precipitation estimation from remotely sensed information using artificial neural networks cloud classification system (PERSIANN-CCS), which extracted cloud features from IR data and transformed them into precipitation through artificial neural networks. Wang et al. [13] constructed a two-staged model for IR precipitation estimation using convolutional neural network (CNN) (IPEC) based on residual blocks as well as a fine-designed five-channel input, which produces better precipitation estimates than PERSIANN-CCS.

For PMW data, Liu et al. [14] designed a multitask framework using multilayer perceptron (MLP) and CNN for simultaneous precipitation identification and estimation, and also verified the feasibility of transfer learning for precipitation retrieval. Orescanin et al. [30] adopted a Bayesian form of residual network (ResNet) architectures to extract the information from PMW data to identify differences in cloud systems while providing precipitation intensity classification. For PR reflectivity data, Chen et al. [15] designed a two-stage deep learning system to link pointwise rain gauge measurements, ground-based radar, and spaceborne radar reflectivity observations. In terms of integrated retrieval, the methods can be categorized into three types based on the input data: integration of multiple SPPs, integration of SPPs, and satellite observations, as well as integration of multimodal satellite observations.

Integration of multiple SPPs refers to use deep learning to merge the precipitation rate of multiple SPPs for bias correction. Wang et al. [31] used a CNN model to merge CMORPH and IMERG with ground-based precipitation products as the target. Gavahi et al. [32] proposed a precipitation data fusion network (PDFN) consisting of 3D-CNN and ConvLSTM to exploit spatial and temporal patterns of precipitation and merge multiple SPPs with rain gauge data as the target. Wang et al. [33] proposed PrecipGAN, which merges incomplete PMW-based precipitation estimates using a generative adversarial network (GAN) conditioned on complete IR-based precipitation estimates.

Integration of SPPs and satellite observations refers to use satellite observations to guide the bias correction of SPP precipitation values. Chen et al. [34] designed an MLP system that uses IR observations to improve PMW-based precipitation estimates. Wu et al. [35] combined CNN and long short-term memory (LSTM) to propose a deep fusion model, which leverages the spatial and temporal correlations in IR data to enhance the quality of TRMM products.

Integration of multimodal satellite observations refers to merge multimodal raw satellite observations for precipitation retrieval. Regarding the integration of IR and PMW, Tang et al. [36] employed a deep neural network (DNN) to estimate high-latitude rain and snow rates, IR can also integrate with lightning data, Yang et al. [37] proposed a deep learning framework that combines GOES-R advanced baseline imager (ABI) and geostationary lightning mapper (GLM) to enhance precipitation retrieval. Yan et al. [38] incorporated attention modules and physical constraints into DNN to retrieve snowfall rates. Gorooh et al. [39] proposed a UNet-based algorithm named Deep-STEP for high spatial resolution precipitation estimation. Regarding the integration of PMW and PR, Arulraj and Barros [40] merged GPM microwave imager (GMI) and Ku-band radar observations using a framework consisting of CNN and MLP for precipitation intensity classification.

However, the first and second types of methods have the possibility to introduce and accumulate the system biases of multiple SPPs, and the third type of methods is limited by overly simple data fusion methods such as concatenating data in the channel dimension. Therefore, we use raw multimodal satellite observations to put an end to the introduction of biases already inherent in SPPs and design an elaborate way to fuse them for effective data integration.

C. Using Geographic Information in Precipitation Retrieval

Satellite precipitation retrieval is susceptible to different topographic and climatic conditions. Consequently, geographical information serves as an additional input or a constraint in many deep-learning methods, Chen et al. [41] designed a CNN to extract complex topographical features from elevation data of the digital elevation model (DEM), quantify the uncertainties, and reduce the biases in CMORPH over complex terrains. Yang et al. [42] developed a bidirectional LSTM (Bi-LSTM) recurrent neural network framework, incorporating elevation, distance to the coastline, and daily maximum and minimum temperatures to correct IMERG. Nguyen et al. [43] improved

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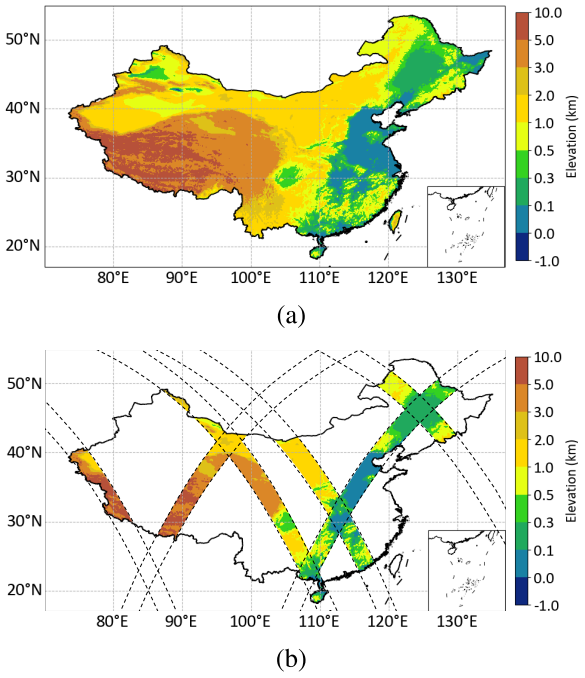


Fig. 2. (a) Elevation of study area. (b) Swaths of DPR through the study area on January 1, 2022.

precipitation estimation in South Korea by merging various resolutions of multiple SPPs with data on Euclidean distance and DEM.

However, most methods simply interact geographical information with the precipitation rates of SPPs. We focus on the deep interaction of geographical information and satellite observations at the feature level.

III. Study Area and Data

A. Study Area

The area considered in this study is the land region of China within the specific range of 73°E–137°E, 18°N–55°N. It should be noted that precipitation over the ocean is not considered. The elevation of the DEM is shown in Fig. 2(a). The region contains a wide variety of terrain, which allows this study to better focus on the effects of complex terrain on precipitation retrieval. More specifically, because of the limited observational range of LEO satellite, we study the regions where the swaths of the DPR through the land region of China. Swath refers to the area of the Earth's surface imaged by a satellite sensor as the satellite revolves around the Earth, as shown in Fig. 2(b).

B. IR Data From FY-4A

FY-4A is the new generation of GEO meteorological satellite of China, launched in 2016 and located over the equator at 104.7°E. The Advanced Geostationary Radiation Imager (AGRI) carried by the FY-4A satellite has 14 channels in the visible and IR bands, with spatial resolutions around 0.5 and 4 km [44]. The AGRI scans a full-disk image in 15 min. Specific spectral bands and main detection objects are shown in Table I. In this study, we use the BTs of channels 7–14, which contain water vapor and cloud-top BT information.

TABLE I
FY-4A AGRI Spectral Band and Main Application

Channel Number	Channel Type	Spectral Bandwidth/μm	Main Application
1	Visible and Near Infrared	0.45–0.49	Small particle aerosol
2		0.55–0.75	Fog, Cloud
3		0.75–0.90	Vegetation
4	Short-wave Infrared	1.36–1.39	Cirrus
5		1.58–1.64	Cloud,Snow
6		2.1–2.35	Cirruis,Aerosol
7	Middle-wave Infrared	3.5–4.0 (high)	Clouds, Fire
8		3.5–4.0 (low)	Land surface
9	Water Vapor	5.8–6.7	High Water Vapor
10		6.9–7.3	Middle Water Vapor
11	Long-wave Infrared	8.0–9.0	Total Water Vapor
12		10.3–11.3	Clouds, surface temperature
13		11.5–12.5	Clouds, total water vapor
14		13.2–13.8	Cloud, water vapor

C. PMW Data From GMI

The GPM Core Observatory (GPM-CO) satellite is an LEO satellite launched in 2014. One of the main payloads carried by the satellite is GMI, a conical-scanning PMW radiometer with a swath of 904 km that provides BTs at 13 channels and both vertical (V) and horizontal (H) polarizations: 10.65V/H, 18.7V/H, 23.8V, 36.6V/H, 89V/H, 166V/H, 183.3 ± 3V, and 183.3 ± 7V GHz. The spatial resolution of GMI's field of view (FOV) varies from 32.1 × 19.4 to 5.8 × 3.8 km as the frequency increases [45]. The BTs in the GMI L1CR (version 07) product [46], which are used in this study, have been recalibrated and remapped to a common basis to resolve FOV differences between the high- and low-frequency channels.

D. Radar Reflectivity Factor From DPR

The other main payload of the GPM-CO is called the dual-frequency PR (DPR). The DPR consists of a Ku-band PR (KuPR) and a Ka-band PR (KaPR) at 13.6 and 35.5 GHz, providing a 5-km resolution footprint with a swath of 245 km. Data collected from the KuPR and KaPR provide 3-D observations of precipitation; in particular, the radar reflectivity factor can reflect the distribution of precipitation particles. Therefore, zFactorMeasured (Z_m) in the DPR L2A (version 07) product [47] is used in this study, which is defined as the measured reflectivity factor without attenuation corrections and calculated using the radar equation from the Ku-band and Ka-band radar echoes, respectively. Both Ku's Z_m and Ka's Z_m have 176 channels of data with a vertical resolution of 125 m, observing precipitation at elevations from 0 to 22 km. In addition, latitude, longitude, and elevation in the DPR L2A product are used as the geographical information in this study.

E. Precipitation Products

1) CLDAS-V2.0: The GT used in this study is the precipitation product in CLDAS-V2.0 product. The CLDAS-V2.0 product is a grid fusion analysis product covering Asian region (0°N–65°N, 60°E–160°E) with a resolution of 0.0625°

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TABLE II
SUMMARY OF THE DATA INFORMATION

Data Type	Name	Sources	Number of Channels	Temporal Resolution/h	Spatial Resolution	Observation Period
Satellite Data	IR	FY4A-AGRI	8	0.25	4km	2018.3 - 2024.3
	PMW	GPM-GMI	13	1.5	13km	2014.3 - present
	Zm	GPM-DPR	176*2	1.5	5km	2014.3 - present
Geographic Information	Elevation	GPM-DPR	1	-	5km	2014.3 - present
	Longitude	GPM-DPR	1	-	5km	2014.3 - present
	Latitude	GPM-DPR	1	-	5km	2014.3 - present
Product	GPM 2B-CMB	-	1	1.5	5km	2014.3 - present
	CLDAS-V2.0	-	1	1	0.0625°	2017.1 - present

and 1 h, including atmospheric driving field products (2-m air temperature, 2-m specific humidity, 10-m wind speed, ground pressure, precipitation, and shortwave radiation) and soil temperature analysis products.

Specifically, the precipitation product of CLDAS-V2.0 is generated by merging SPPs (FY-2/CMORPH) with ground-based observation data from over 2400 quality-controlled national automatic weather stations and nearly 40 000 regional automatic weather stations [48], [49]. The merging process [50] is primarily driven by ground-based observations, with satellite data adjusted via optimum interpolation (OI) [51] and probability density function (pdf) matching methods [52]. Benefiting from abundant and high-quality ground-based observations, as well as reliable data merging approaches, the precipitation product of CLDAS-V2.0 exhibits high accuracy over China. Numerous validation studies have demonstrated its superiority compared to international datasets [53], [54], [55], [56]. Therefore, it is selected as the GT in this study.

2) GPM 2B-CMB: The baseline for performance comparison with our model is surface precipitation rate (*surfPrecipTotRate* variable) in GPM Level-2B DPR and GMI Combined (2B-CMB) product (version 07) [57]. The 2B-CMB product combines DPR reflectivity and GMI BTs to provide, in principle, the most accurate surface precipitation and vertical distribution of precipitation with a resolution of 5 km and 1.5 h can be achieved from a spaceborne platform. Therefore, it is a common reference for cross-calibration of other PMW precipitation estimates in the GPM constellation.

F. Database Construction

This study proposes a large-scale precipitation retrieval database based on multimodal satellite data called Precipitation-MG, consisting of 3886 satellite swaths collected from 2021 to 2022 with a spatial resolution of 5 km. Each swath includes three types of satellite data, geographical information, and two precipitation products, mentioned above. A summary of the data information is shown in Table II. It is worth noting that our database comprises the comprehensive satellite observations currently utilized for precipitation retrieval, aimed at fundamentally exploring the interaction among these observations and establishing the relationship between satellite observations and precipitation. The details of data processing are outlined as follows and shown in Fig. 3.

1) Date Split: The swath passed over this study area thousands of times over the two-year period, following the exclusion of very brief swaths or swaths containing invalid values, we collected a total of 3886 times (1935 times in 2021 and 1951 times in 2022). The precipitation pattern in China shows the variability apparently across different seasons. Therefore, the data of 1935 swaths in 2021 are split into a training set and a validation set with relative sample sizes of 0.8 and 0.2 in each month. For a more complete test of model's year-round continuous performance, the data of 1951 swaths in 2022 are used entirely as a testing set.

2) Temporal and Spatial Matching: We used the temporal and spatial information from DPR as a reference for processing data. The DPR has a spatial resolution of 5 km and a temporal resolution of 1.5 h. Since we only select data that pass over the study area (approximately four–six times per day), the temporal resolution of our dataset is roughly 4 h. Briefly, our dataset has a 5-km spatial resolution and a roughly 4-h temporal resolution. Temporally, GMI BTs, DPR Z_m, 2B-CMB precipitation, and the geographical information are sourced from GPM-CO, ensuring their alignment. Thus, we only need to find and match the corresponding FY-4A BTs and CLDAS-V2.0 precipitation. Based on the start time of the DPR swath entry into the study area, we find the FY-4A BTs that are closest to this time, as well as the CLDAS-V2.0 precipitation recording the accumulated precipitation within that hour (e.g., the 01:00 CLDAS-V2.0 records the accumulated precipitation from 00:00 to 01:00). Spatially, we use the linear interpolation method to match GMI BTs, FY-4A BTs, and CLDAS-V2.0 precipitation into 5-km swath pixels, aligned with the rest data.

3) Swath Crop: For the training set, since each swath has the same width of 49 pixels but has different lengths, we need to crop the data into patches of the same size to ensure batch training; therefore, we use a sliding window of 49 × 256 pixels to crop the data, resulting in patches of 49 × 256 pixels with N bands. The sliding step of cropping is set to 128 to facilitate data augmentation by resampling. For easier cropping, the initial data are padded with zeros at the end to ensure that the length is an integer multiple of 128. However, for the validation and testing set, we retain the original data size (width 49 × length L × bands N) of the swaths because if the model predicts using the data in patch form and splices the predicted results, significant errors are likely to occur at the boundaries.

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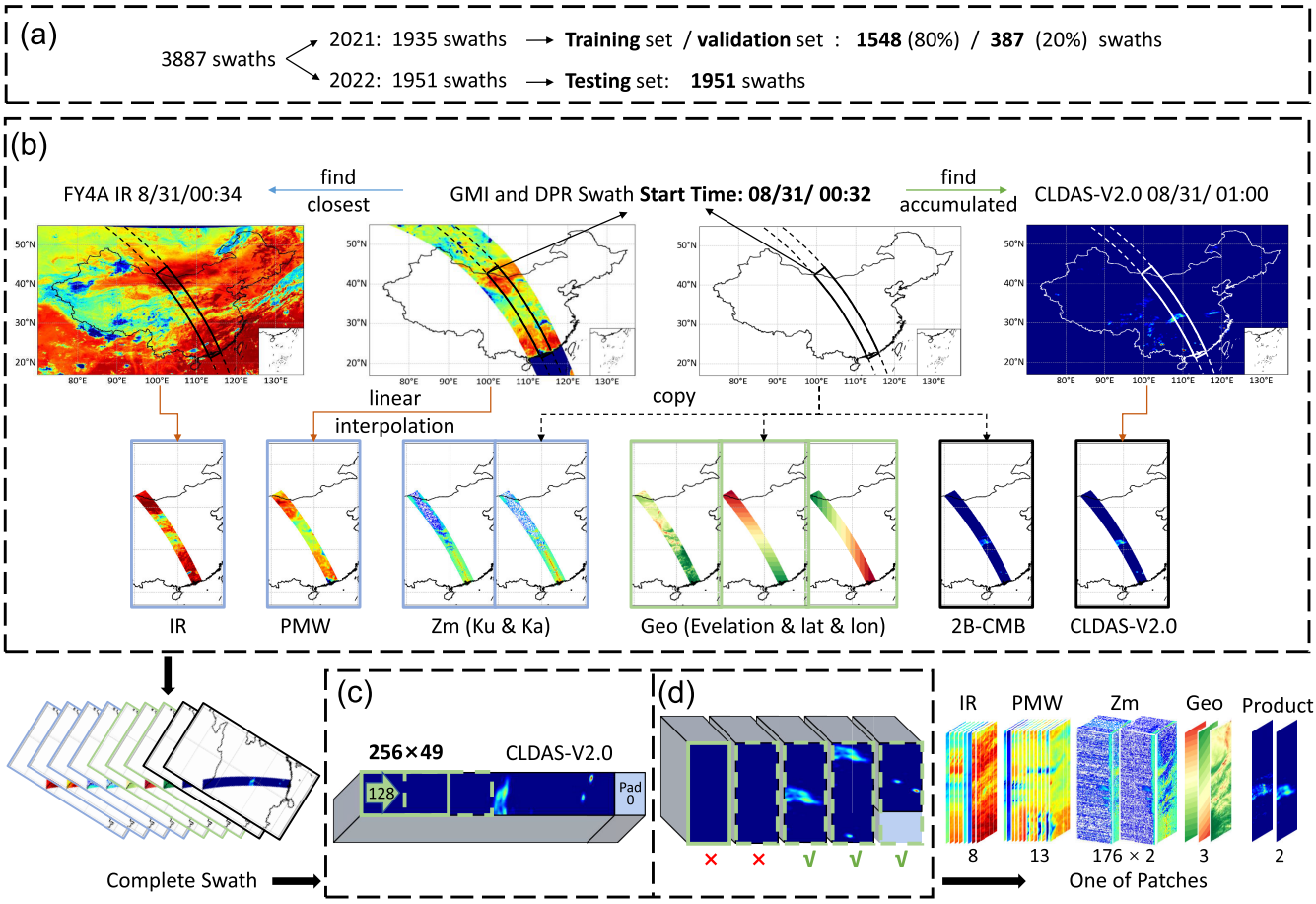


Fig. 3. Data preprocessing methods, including (a) data split, (b) temporal and spatial matching, (c) swath crop, and (d) sample balance. Validation set and testing only need to go through step (b) to keep the shape of the complete swath, while the training set should go through all steps to create uniform patches.

4) Sample Balancing: To mitigate the sample imbalance between rain and nonrain, patches without precipitation pixels are excluded from the training set, and the precipitation threshold is set to 0.1 mm/h (i.e., precipitation rate $\geq$ 0.1 mm/h are defined as precipitation event). Nevertheless, in order to test the model's performance comprehensively, we do not perform sample balancing in the validation and testing sets.

After completing the processing steps, there are 2281 patches for training, 387 swaths for validation, and 1951 swaths for testing.

IV. Methodology

A. Overall Framework

The overall framework is shown in Fig. 4. The precipitation retrieval task is divided into two primary tasks: precipitation/no precipitation (P/NP) identification and precipitation rate regression. We propose a multimodal precipitation retrieval UNet (MPR-Unet) that is capable of handling both tasks by only changing the number of output channels (2 and 1) in the last convolutional layer. The network responsible for identification is called MPR-Unet-ide, while the one responsible for regression is called MPR-Unet-reg.

The two models are trained independently in the training phase, using different functions to calculate the losses between predicted results and target values provided by the CLDAS-V2.0. MPR-Unet-ide uses the cross-entropy (CE) loss, and the MPR-Unet-reg uses the mean-squared error (mse) loss. In the validation and testing phases, the final precipitation retrieval result is obtained by fusing the results from two networks. Specifically, MPR-Unet-reg results are continuous precipitation rates, while MPR-Unet-ide results are P/NP masks consisting of 1 and 0. We combine them by elementwise product, ensuring that only pixels identified as precipitation by MPR-Unet-ide are assigned the corresponding precipitation values in the MPR-Unet-reg result.

As for the input data, the IR and PMW consist of 8 and 13 channels, respectively, while the terrain data have three channels. The radar reflectivity factor data have 352 channels data concatenating Ku- and Ka-bands, we downsample it to 176 channels by retaining the odd-indexed channels to reduce data redundancy, and the vertical resolution decreases from 125 to 250 m. Consequently, the original 200-channel data stacked with three types of satellite data and the geographical information serve as input to the model. In the training phase, the input data size for the feature model is $49 \times 256 \times 200$, while during the validation and testing phases, the input data size is $49 \times L \times 200$.

B. Architecture of MPR-Unet

As illustrated in Fig. 4, the network architecture is built based on the UNet. We also designed an MBF module and a GIC module in MPR-Unet.

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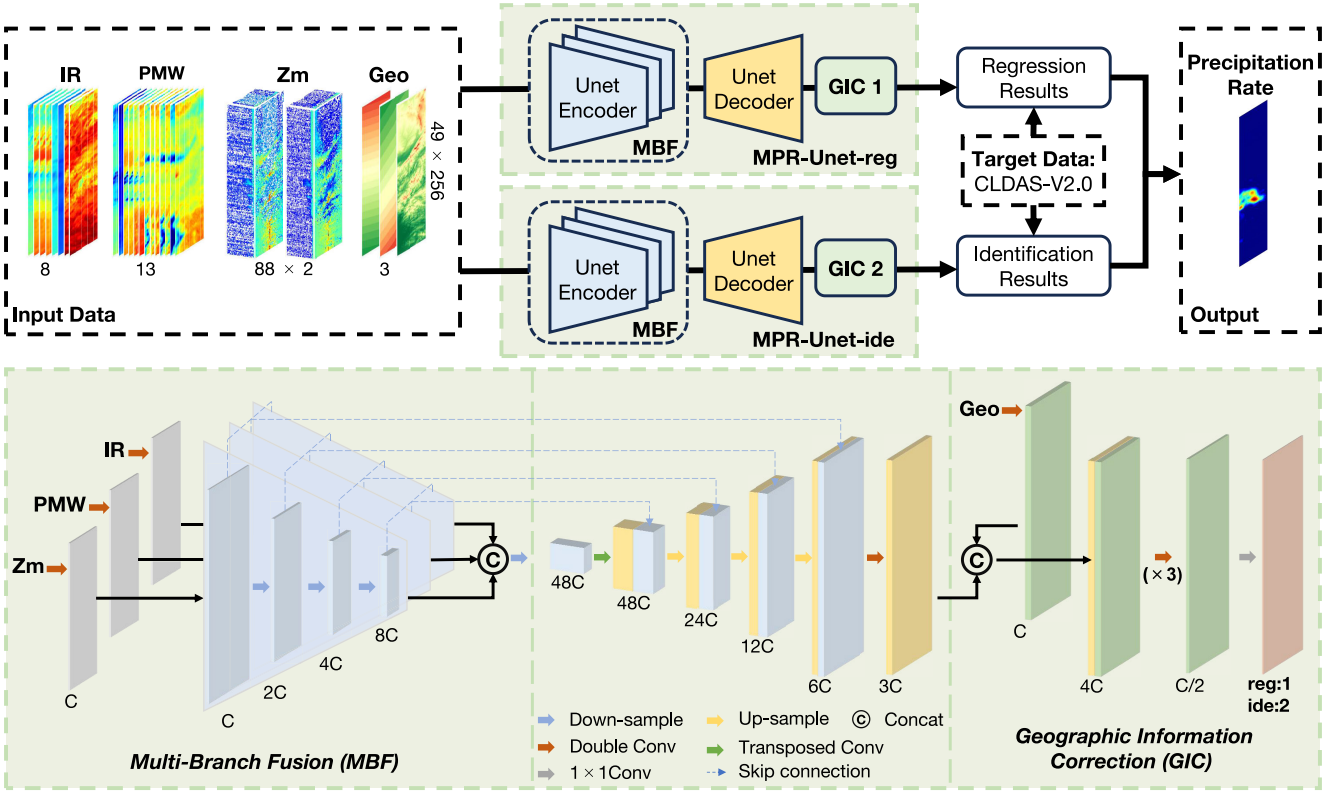


Fig. 4. Framework of PRMG. The final retrieval result is obtained by fusing the results from two networks. The C in architecture of networks represents the number of feature layers, $C = 32$.

1) UNet Configuration and Setting: The UNet has been widely used in medical segmentation [58], [59], [60] and has shown great potential in precipitation retrieval [2], [39], [61]. As a fully convolutional network, it possesses the capability to flexibly accept inputs of various sizes. This adaptability is particularly advantageous for our precipitation retrieval task to effectively handle orbit data of varying lengths. We use three UNet encoders and a UNet decoder in our architecture. In the encoder, the downsampling operation is composed of a double-conv block and a 2×2 max-pooling operation, the double-conv block has two 3×3 convolutions, and each convolution is followed by a batch normalization (BN) module and an activation function called rectified linear unit (ReLU). In the decoder, the upsampling operation is composed of a 2×2 transpose convolution and a double-conv block. We initial the number of first feature layers to C (i.e., 32), and each downsampling operation doubles the number of feature channels, while each upsampling operation halves it.

2) MBF Module: The three types of satellite data have distinct characteristics but similar semantic information related to precipitation. The fusion of multimodal data can combine complementary information and integrate the strengths among different modalities to enhance the performance of precipitation retrieval models. Therefore, we utilize multibranch architecture to extract and fuse data features focusing on the two critical aspects: the fusion stage and the fusion method. As illustrated in Fig. 5, the fusion stage can be categorized into early, middle, and late fusion, while the fusion method can be categorized as addition (add), concatenate (cat), and multiply (mul). In Section V-B, we report on nine fusion strategies and conduct analyses. Ultimately, we choose the middle stage with the cat method to design this MBF module, as shown in Fig. 4. The middle stage features have an optimal degree of abstraction and contain both rich semantic information and sufficient detail. By fusing these features with the cat method, the model can retain and fully utilize these features, minimizing information loss and conflict. This fusion strategy also ensures better interaction and optimization of features within the decoder. Specifically, three types of satellite data, each with a different number of channels, are uniformly transformed into three 32-layer features by a double-conv block and processed separately by the encoder in each branch. Then, we use a cat operation to aggregate the outputs of three encoders and downsample, subsequently feeding them into the UNet decoder. The skip-connecting operation concatenates the downsampling features in the three encoders with the corresponding upsampling feature in the decoder.

3) GIC Module: The complex terrain of different regions has a significant impact on precipitation, often resulting in abnormal weather conditions. SPPs often exhibit underestimation or overestimation in regions with complex terrain. Furthermore, different latitudinal and longitudinal regions have distinct climatic characteristics, resulting in region-specific precipitation patterns. With the assistance of geographical information, the model can quantify and correct the uncertainties caused by terrain and climate in precipitation retrieval, resulting in more accurate retrieval results. In previous studies, some research extracted the geographic features and fused

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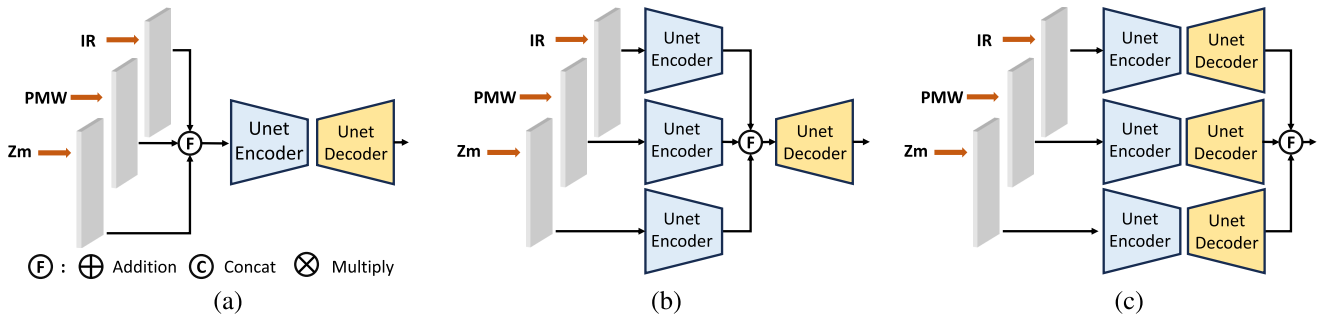


Fig. 5. Illustrations of different fusion strategies. Each fusion stage can be coupled with different fusion methods, resulting in nine kinds of fusion strategies. (a) Early stage. (b) Middle stage. (c) Late stage.

them with satellite observation features to retrieve precipitation, while others utilized the geographical information to guide the model in correcting the final precipitation results of SPPs. Since satellite observation features contain more information than the final precipitation rates, the former approach fuses geographic features with satellite observation features and enables richer information complementarity and deeper multimodal interaction. This enhances the model's generalization capability and improves its ability to correct precipitation retrieval biases in complex terrains. In contrast, the latter approach does not incorporate geographic information into the core retrieval process from observational data to precipitation and may rely on specific statistical relationships, limiting its generalization across different regions and climatic conditions. Therefore, we follow the former approach. Besides, compared to the dynamic data captured by satellite observations in real time, the geographical information is static data that has no direct semantic correlation with precipitation [21]. Fusing them at an early stage may introduce additional noise into the satellite observation features. As a result, we fuse them at a late stage. Specifically, the geographical information containing elevation, longitude, and latitude as three-channel data is extracted into a 32-channel geographic feature by the double-conv block. The geographic features are concatenated with the satellite features from the decoder output of the UNet. Three double-conv blocks are applied to facilitate the interaction between these features. Finally, a 1×1 convolution layer, which has two output channels in MPR-Unet-ide and one output channel in MPR-Unet-reg, is used to map the feature vector to the target output.

C. Evaluation Metrics

We evaluate the performance of precipitation classification and precipitation rate regression by two kinds of evaluation metrics, respectively. Identification metrics are used to evaluate precipitation identification performance, including the probability of detection (POD), false alarm rate (FAR), and critical success index (CSI), which are given as follows:

$$\mathrm{POD} = \frac{H}{H+M} \tag{1}$$

$$\mathrm{FAR} = \frac{F}{H+F} \tag{2}$$

$$\mathrm{CSI} = \frac{H}{H+M+F} \tag{3}$$

where H (hit) is the number of pixels predicted by the model and labeled by CLDAS-V2.0 as precipitation; M (miss) is the number of pixels labeled as precipitation but predicted as no precipitation; and F (false alarm) is the number of pixels predicted as precipitation but labeled as no precipitation.

Regression metrics are used to evaluate precipitation estimation performance, including the root-mean-square error (RMSE) and the Pearson correlation coefficient (CC), which are given as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2} \tag{4}$$

$$\mathrm{CC} = \frac{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sigma_{\hat{y}}\sigma_y} \tag{5}$$

where $\hat{y}$ is the model prediction; y is the CLDAS-V2.0 precipitation; n is the number of all samples; $\bar{\hat{y}}$ is the averaged model precipitation estimate; $\bar{y}$ is the averaged CLDAS-V2.0 precipitation; $\sigma_{\hat{y}}$ is the standard deviations of model precipitation series; and σ_y is the standard deviations of CLDAS-V2.0 precipitation.

V. Experimental Results and Analysis

A. Implementation Details

All the experiments are implemented on the platform with one NVIDIA RTX A6000 GPU. The training batch size is set to 64, and the training epoch is set to 200, all models are trained using the root-mean-square prop (RMSProp) optimizer with a starting learning rate of $1e^{-5}$ and decaying to one-half per 30 epochs. In addition, before being applied to the deep learning model, the original 200-channel data contain the satellite data and the geographical information should be normalized between 0 and 1 as follows:

$$X_{i,\mathrm{norm}} = \frac{X_i - X_{i,\min} + X_{i,\mathrm{offset}}}{X_{i,\max} - X_{i,\min}} \tag{6}$$

where $X_{i,\max}$, $X_{i,\min}$, and $X_{i,\mathrm{offset}}$ are the maximum, minimum, and offset values used to normalize the channel i in the original data; these values of different types of data are presented in Table III. All the experimental results shown below are obtained by testing on the 2022 test set.

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TABLE III
MAXIMUM AND MINIMUM VALUES USED FOR NORMALIZING THE INPUT DATA

Type	Maximum	Minimum	Offset
IR(k)	500	100	0
PWM(k)	350	50	0
Z_m(dbz)	100	0	0
Latitude(°)	180	0	90
Longitude(°)	360	0	180
Elevation(m)	10000	0	0

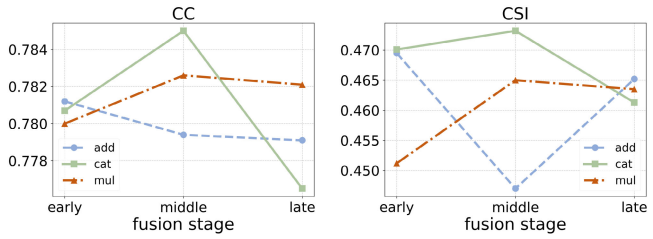


Fig. 6. Performance of different fusion strategies.

B. Ablations of Fusion Strategy

We aim to explore different fusion strategies of three satellite modalities to effectively exploit the advantages of each modality to achieve better precipitation retrieval results. We mainly consider the fusion strategy from two perspectives, i.e., the fusion stage and the fusion method. Combining three fusion stages (early, middle, and late) with three fusion methods (add, cat, and mul), we evaluate a total of nine architectures, as shown in Fig. 5. The MPR-Unet-ide and MPR-Unet-reg model architectures are adjusted synchronously. And to ensure a purely comparison of different fusion strategies, the GIC module is removed during ablation experiments, as it introduces additional effects beyond feature fusion. Apart from the output layer, the architectures of the two models remain consistent.

The performance of different fusion strategies on an hourly time scale is reported in Fig. 6, mainly focusing on the CC for regression performance and the CSI for identification performance. Regarding the three kinds of fusion stages, the performance shows a trend of increasing and then decreasing, except for the add method. For instance, using the cat method, the CC increases from 0.7807 to 0.7850 and then decreases to 0.7765, and the CSI increases from 0.4701 to 0.4732 and then decreases to 0.4613. This trend indicates that the middle fusion of the satellite data has a greater potential. This can be attributed to the following reasons.

1) In the early stage, the data have not been processed into high-level features. Early fusion may result in the information being too mixed to extract valid features and complementary information.
2) In the late stage, the features are highly abstract. Although they carry high semantic information, some details from the original data might be lost. The network processes each modality separately for most of the learning process, which limits the ability of the network to learn complementarity between the modalities.
3) In the middle stage, where the data have undergone some feature extraction. Middle fusion achieves a balance between abstraction and detailed retention and ensures a balance between the independent learning and integrated learning of the network. These balances allow the network to perform thorough feature extraction and interaction, better exploit the strengths of each modality, and enhance complementary relationships.

Regarding the three kinds of fusion methods, the compatibility of the three fusion methods varies at different stages. Regarding CC in the early stage, the add, cat, and mul methods show similar performance with CC of 0.7812, 0.7807, and 0.7800, respectively. In the middle stage, the cat outperforms both add and mul with 0.7850. However, in the late stage, mul is 0.7821, better than add, while cat drops to the lowest. A similar phenomenon is observed in the CSI. This indicates that the effectiveness of the fusion method is influenced by the fusion stage. Taking the performance of add, cat, and mul in the middle stage as an example, the reasons contributing to this phenomenon are as follows. The features in the mid stage have higher level semantic information. However, due to the independent extraction of the three modalities, there are different ranges of value distribution and some associated noise in each feature.

1) The add method can lead to conflicts and cancellation of information between different features, especially when the features have different magnitudes and scales. Important information may be weakened or lost, and noise accumulation can also occur.
2) The mul method can easily lead to excessively small or large feature values, resulting in the distortion of important information. Additionally, the nonlinear characteristics of the mul method can lead to gradient vanishing or explosion problems during the training process, making the training unstable.
3) The cat method preserves the original information of all features, avoiding information loss. It also distinguishes the characteristics from different data sources, allowing the model to flexibly learn the relationships of these features in subsequent layers. This method retains more information and diversity, contributing to improved model performance. Therefore, the cat method is more suitable for feature fusion in the middle stage, achieving the best performance in both identification and regression.

Overall, the model architecture combining middle fusion and the cat method, benefiting from their respective advantages and mutual compatibility, achieved the best performance among the nine architectures. We also conducted experiments with the GIC module included; although the results show some variations, this combination still achieves the best performance, further demonstrating the stability of this fusion strategy. Therefore, we consider this combination to be the most suitable architecture for the fusion of the three types of satellite data in this study, and we eventually used this architecture in MBF of MPR-Unet-ide and MPR-Unet-reg.

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TABLE IV
PERFORMANCE OF HOURLY PRECIPITATION RETRIEVAL FROM DIFFERENT INPUTS

IR	PMW	Z_m	RMSE(mm/h) ↓	CC ↑	POD ↑	FAR ↓	CSI ↑
✓			0.5239	0.3080	0.1228	0.5473	0.1390
	✓		0.3774	0.7264	0.4794	0.2611	0.4099
		✓	0.3476	0.7738	0.5391	0.2560	0.4547
✓	✓		0.3745	0.7337	0.4991	0.2775	0.4188
✓		✓	0.3458	0.7787	0.5296	0.2560	0.4480
	✓	✓	0.3461	0.7805	0.5391	**0.2327**	0.4633
✓	✓	✓	**0.3414**	**0.7850**	**0.5583**	0.2435	**0.4732**

TABLE V
PERFORMANCE OF HOURLY PRECIPITATION RETRIEVAL FROM COMBINATIONS OF DIFFERENT MODULES

MBF	GIC	RMSE(mm/h) ↓	CC ↑	POD ↑	FAR ↓	CSI ↑
×	×	0.3437	0.7798	0.5399	0.2323	0.4641
✓	×	0.3414	0.7850	0.5583	0.2435	0.4732
×	✓	0.3379	0.7889	0.5553	0.2442	0.4708
✓	✓	**0.3367**	**0.7918**	**0.5594**	**0.2400**	**0.4754**

TABLE VI
PERFORMANCE OF HOURLY PRECIPITATION RETRIEVAL FROM DIFFERENT ELEVATION RANGES

	RMSE(mm/h) ↓		CC ↑		CSI ↑	
Elevation Range	w/o GIC	w/ GIC	w/o GIC	w/ GIC	w/o GIC	w/ GIC
<2000m	0.3715	**0.3694**	0.8058	**0.8093**	**0.5286**	0.5270
2000m ∼ 4000m	0.2775	**0.2589**	0.6661	**0.6945**	0.3536	**0.3759**
≥4000m	0.2458	**0.2357**	0.4119	**0.4203**	0.2573	**0.2657**

C. Ablations of Input Modalities

To explore the effectiveness of different satellite data fusions in improving the model's precipitation retrieval performance, we first conducted an ablation experiment on the model input. Each type of satellite data is fed into the model through its own branch in the MBF module, so we can control the model input by retaining only the branches that are needed. It should be noted that the GIC module remains removed. There are a total of seven input modes for the three types of data, and the results of the experiment are listed in Table IV.

The result indicates that the fusion of different satellite data is effective in the precipitation retrieval task. From the regression performance reflected by RMSE, CC, and the overall identification performance reflected by CSI, it can be seen that the best result can be achieved when fusing three satellite data; meanwhile, fusing any two data can be better than using these two data alone, except for the CSI of the IR and Z_m fusion compared to the CSI of the Z_m alone.

The model can perform well when using PMW and Z_m data alone due to the strong physical relationship between such data and precipitation. However, the performance of the model is relatively poor when using only IR data because IR-based precipitation retrieval is confined to the indirect estimation relationship between cloud top features and precipitation. Therefore, when multiple data are fused, the inclusion of IR data might sometimes have adverse effects, e.g., the CSI of the model using IR and Z_m is lower than the CSI of the model using Z_m alone. However, it is more noteworthy that data fusion is committed to the complementation and calibration of multiple data, and the deep learning model with adaptive learning capability can filter unfavorable information in the fused data, extract beneficial information, and promote interaction between the information, leading to performance improvements in more cases as demonstrated by our experiments. In particular, when fusing all three data for precipitation retrieval, the comparative results show that each data contributes to performance improvement.

D. Ablations of Module Components

To evaluate the effectiveness of the proposed two modules, we conduct ablation experiments by adding different modules to the UNet architecture. The results of classical UNet, PRMG w/o GIC, PRMG w/o MBF, and full PRMG are listed in Table V. In classic UNet, the three satellite data are concatenated and input to the UNet architecture. Without either module, it achieves the lowest performance on all metrics. Adding either the MBF or GIC module improves the performance compared to classic UNet. In terms of identification metrics, MBF shows more improvement than GIC, while in terms of regression metrics, GIC performs better than MBF. When MBF and GIC are used together in PRMG, they yield the best performance.

As analyzed in Section IV-B, the MBF module significantly improves model performance due to its utilization of middle

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TABLE VII
PERFORMANCE COMPARED TO OUR METHOD WITH TRADITIONAL AND DEEP LEARNING METHODS

	Backbone	RMSE (mm/h)↓	CC↑	POD↑	FAR↓	CSI↑
GPM 2B-CMB [57]	Traditional	0.5953	0.4511	0.4241	0.4367	0.3192
PCM [40]	1D CNN	0.4076	0.6701	**0.7149**	0.6502	0.3070
ABI-GLM [37]	2D CNN	0.4529	0.5915	0.6726	0.8840	0.1098
Deep-STEP [39]	UNet	0.3632	0.7508	0.4290	**0.2203**	0.3826
Swin-Unet [60]	UNet,Transformer	0.3692	0.7404	0.4709	0.2807	0.3978
PRMG(ours)	UNet	**0.3367**	**0.7918**	0.5594	0.2400	**0.4754**
PRMG w/o GIC	UNet	0.3414	0.7850	0.5583	0.2435	0.4732

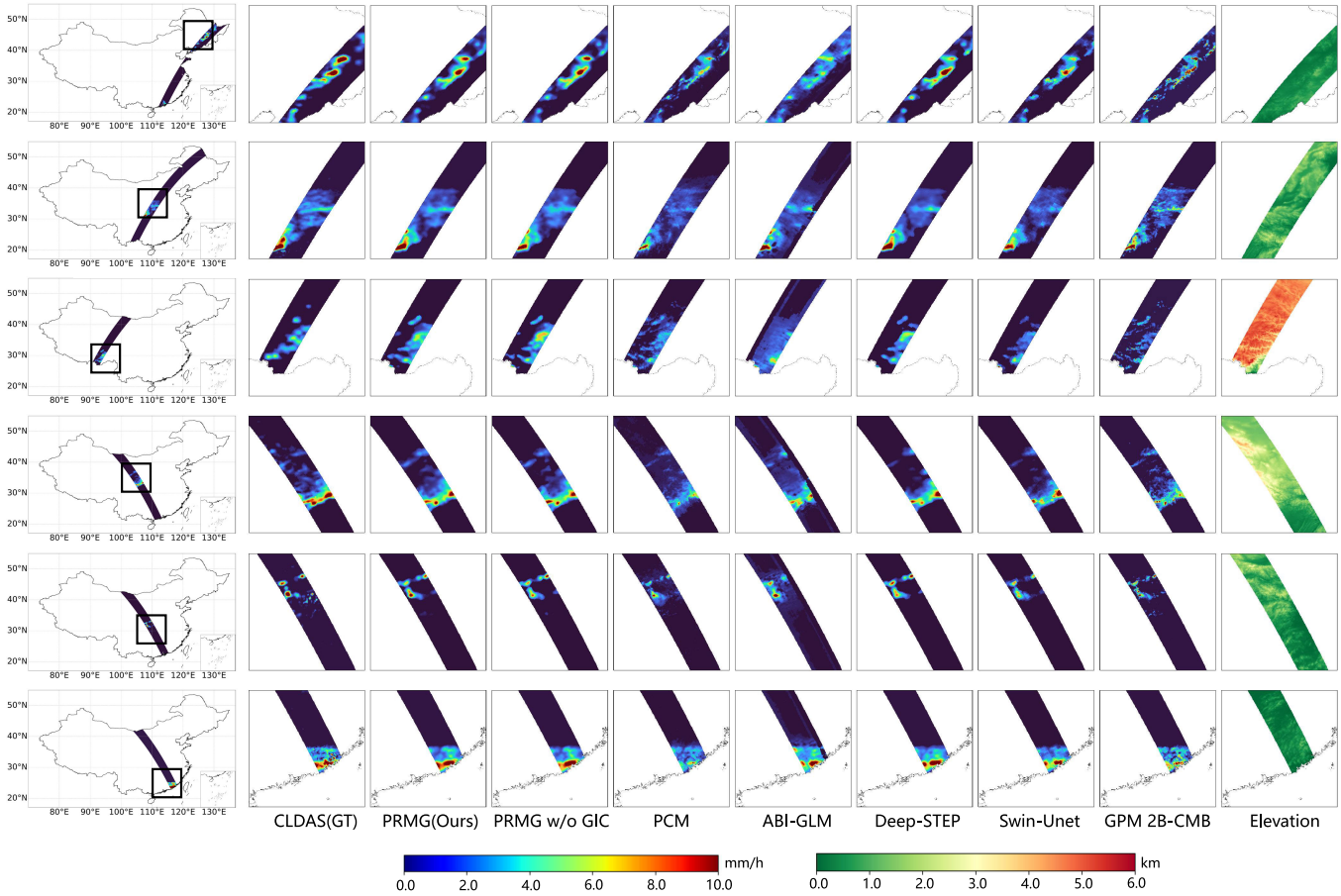


Fig. 7. Illustrations of different hourly precipitation results of traditional and deep learning methods.

fusion and the concatenate method, which strengthens data interaction and better integrates the three types of satellite data. Therefore, the inclusion of MBF in the model results in performance improvements. Concurrently, the GIC quantifies the impact of geographical information on precipitation retrieval and establishes a correction relationship for satellite-derived precipitation based on geographical information. Thus, incorporating GIC enhances model performance.

To further evaluate GIC, we compare the performance of PRMG w/o GIC and the full PRMG in different elevation ranges in terms of CC and CSI. The results are presented in Table VI. Across the three altitude ranges, the improvement is more significant at elevations above 2000 m compared to those below. This characteristic enables our model to effectively serve precipitation retrieval tasks in high-altitude regions; we can reduce the errors in precipitation retrieval caused by the complex terrain.

Overall, these two modules play an effective role in data fusion and consideration of geographical information, respectively, and together can further improve the performance.

E. Performance Comparison

To validate the effectiveness of the proposed PRMG, we compare it with traditional methods (i.e., GPM 2B-CMB), several multimodal precipitation retrieval methods based on deep learning, including PCM [40], ABI-GLM [37], and Deep-STEP [39], along with Swin-Unet [60], an advanced and widely recognized variant of UNet that incorporates Transformer modules [62]. During the training process, all

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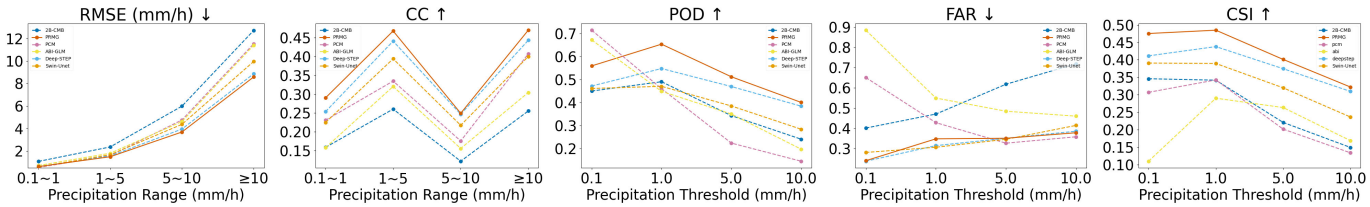


Fig. 8. Quantitative evaluation scores of the methods in different ranges of precipitation. The evaluation scores of RMSE and CC are calculated using the data in different value ranges while POD, FAR, and CSI are calculated with different thresholds using all the data.

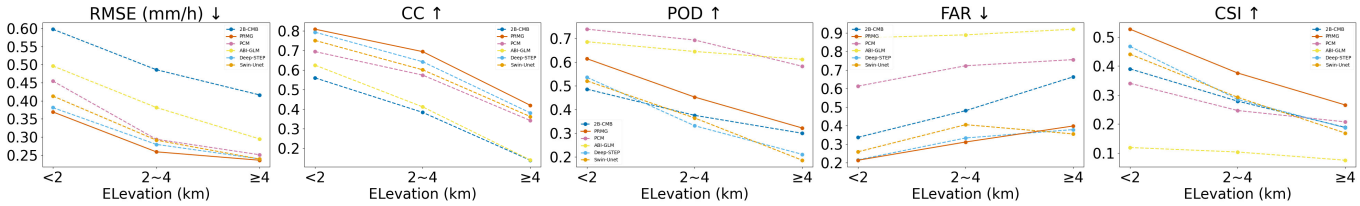


Fig. 9. Quantitative evaluation scores of the methods in different ranges of elevation.

models are trained using the same GT (i.e., CLDAS-V2.0), with consistent input data comprising 197 satellite observation channels and three geographic information channels.

The results are presented in Table VII. For the identification scores, PRMG achieves a CSI of 0.4754, significantly outperforming all compared methods. Specifically, while PCM exhibits a high POD, it struggles with false alarms; Deep-STEP, on the other hand, achieves the lowest FAR but suffers from a low hit rate. PRMG achieves a balance by improving the hit rate while maintaining a low FAR, resulting in the best overall precipitation identification performance. For regression scores, PRMG also shows an advance over the other methods, with an RMSE of 0.3367 and a CC of 0.7918. Based on these comparisons, PRMG demonstrates superior overall performance in both precipitation identification and regression compared to the GPM 2B-CMB product.

Visual comparison of hourly precipitation is provided in Fig. 7, including the CLDAS-V2.0 product as GT, the proposed PRMG method, and the other methods used for comparison. PRMG successfully identifies precipitation areas missed by the other methods and provides precipitation rates that are closer to CLDAS-V2.0. Compared to the PRMG w/o GIC, PRMG exhibits better performance in retrieving both weak and strong precipitation across various terrains.

To evaluate the performance of PRMG and the other methods under different precipitation ranges, we calculate POD, FAR, and CSI at four precipitation thresholds: 0.1, 1, 5, and 10. Additionally, we calculate RMSE and CC across different precipitation ranges defined by these thresholds. The quantitative evaluation scores are illustrated in Fig. 8. It can be found that the performance declines as the thresholds increase. This is expected, as higher precipitation levels are typically associated with more complex and transient meteorological features, which pose greater challenges for satellite-based precipitation retrieval. However, across all precipitation ranges, PRMG consistently demonstrates more reliable performance compared to the other methods, as evidenced by its lower RMSE and higher CC and CSI values.

We also evaluate their performance across different elevation ranges, considering three elevation ranges divided by 2 and 4 km. The quantitative evaluation scores are illustrated in Fig. 9. As elevation increases, there is a corresponding decline in precipitation retrieval performance. However, compared to the other methods, PRMG also consistently demonstrates better performance across all elevation ranges.

Overall, the proposed PRMG outperforms the other methods in both visual and quantitative comparisons, especially for different precipitation ranges and elevation ranges. This demonstrates the effectiveness of our method.

F. Analysis of Precipitation Distribution and Structure

To further evaluate the capability of different methods in capturing the structural characteristics and distribution patterns of precipitation, we assess the pdf and power spectral density (PSD) of the precipitation rates.

1) PDF Analysis: The pdf describes the statistical distribution of precipitation, characterizing the relative likelihood of different precipitation ranges. A method with a pdf close to that of the GT demonstrates a better ability to reflect the diverse intensities and frequencies observed in real precipitation events. Fig. 10 shows the pdf of precipitation rates in four different ranges: light (0.1–1 mm/h), moderate (1–5 mm/h), heavy (5–10 mm/h), and extreme (10–20 mm/h). To prevent the probability density of low precipitation values (0–0.1 mm/h) from dominating the distribution, we first select the pixels where CLDAS-V2.0 indicates precipitation ($\geq$0.1 mm/h). Based on these pixels, the PDFs for each method are computed using their respective precipitation retrieval result.

In the light precipitation range, due to the exclusion of zero or near-zero precipitation values as explained above, the CLDAS-V2.0 as GT exhibits a prominent peak near 0.45 mm/h rather than at 0 mm/h. In contrast, the peak positions of all comparative methods appear at lower precipitation rates, indicating a systematic tendency to excessive underestimation of precipitation. Among the evaluated methods, PRMG,

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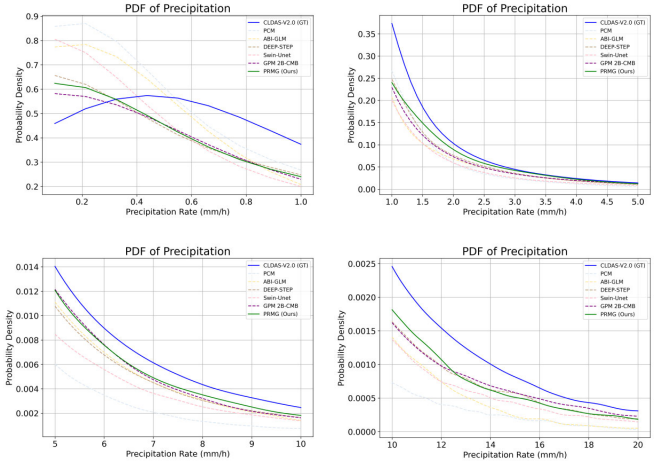


Fig. 10. PDF of precipitation rates.

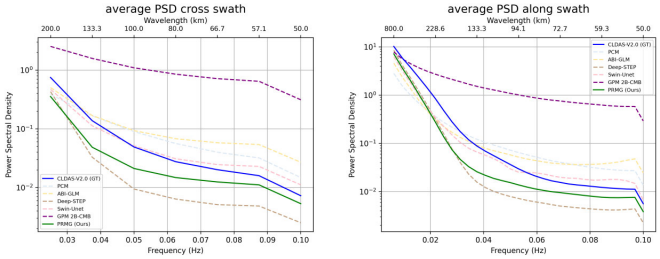


Fig. 11. PSD of precipitation rates.

Deep-STEP, and GPM 2B-CMB exhibit probability densities that more closely follow the shape and trend of the CLDAS-V2.0, whereas PCM, ABI, and Swin-Unet display a steeper decline after the peak, suggesting a more pronounced deviation in retrieval light precipitation. In the moderate to heavy precipitation range (110 mm/h), PRMG demonstrates notable superiority, with its probability densities aligning more closely with GT. This suggests a better ability of the model to capture the distribution of precipitation in this range. However, for extreme precipitation events (above 10 mm/h), the probability density of PRMG, Deep-STEP, and GPM tends to converge and deviate similarly from the CLDAS-V2.0. In summary, while PRMG demonstrates superior overall performance, it still exhibits issues with precipitation underestimation and the capture of extreme precipitation, suggesting potential directions for further improvement.

2) PSD Analysis: The PSD characterizes the spatial structure of precipitation by quantifying the distribution of spatial variability across different frequencies. Low frequencies (left side of the plot) correspond to large spatial scales, i.e., broad precipitation systems that extend over large areas. Higher PSD values at low frequencies indicate that a method captures more large-scale variability, which is typically associated with major weather systems. High frequencies (right side of the plot) correspond to small spatial scales, such as local, fragmented precipitation patterns. Higher PSD values at high frequencies may reflect more small-scale variability, but excessively high power in this range can also indicate noise or over-fragmentation, which may not correspond to real physical structures. We evaluating PSD in both along-track and cross-track directions for a more comprehensive assessment of how well a method captures the 2-D organization of precipitation fields. Additionally, it should be clarified that due to the irregular satellite revisit cycle of our data for a specific region, and it is not feasible to calculate the variability at different time scales using PSD. Therefore, only spatial PSD analysis was conducted.

As shown in Fig. 11, CLDAS-V2.0 (i.e., GT) exhibits a gradual decrease in spectral power from large to small scales, which aligns with the real-world spatial variability patterns of precipitation. The proposed PRMG method and the Swin-Unet yield PSD curves that are closer to the GT trend than other methods across both swath directions, suggesting a better ability to learn and reproduce the natural spatial variability of precipitation. In contrast, most methods display relatively high power at smaller scales, which may indicate the presence of spurious small-scale noise or fragmented precipitation patterns. However, the PRMG generally underestimates the overall PSD magnitude compared to the GT, indicating that the total spatial variability is slightly suppressed. Although the spectral slope of the PRMG aligns well with the GT, Swin-Unet achieves a closer match to the GT in terms of absolute PSD values. This suggests that PRMG retains potential for further improvement in representing the spatial variability of precipitation. Overall, this analysis uses PSD to compare PRMG with various methods; although it shows results that are relatively consistent with GT, it retains potential for further improvement in representing the spatial variability of precipitation.

VI. Conclusion

In this article, a novel deep-learning framework of precipitation retrieval named PRMG and a new database called Precipitation-MG are proposed. Via MBF module and GIC module, the multimodal satellite observations and geographic information are effectively utilized by the proposed PRMG. Our method outperforms not only the traditional method used to produce the GPM 2B-CMB product, but also several deep learning-based multimodal precipitation retrieval methods, demonstrating superior performance across multiple evaluation metrics. We also demonstrate the contribution of the module in our framework through a series of ablation experiments, which shows the following.

1) The integration of multimodal satellite observations can be effectively achieved through a multibranch architecture employing middle-stage fusion via concatenation, which significantly enhances the model's accuracy.
2) The introduction of geographical information into precipitation feature construction serves to correct the distortions caused by complex terrain, ultimately enhancing the overall performance of the model.

The proposed Precipitation-MG database can provide data support for the application of the deep learning method in the field of satellite precipitation retrieval and the evaluation of other methods. For future works, we intend to utilize swath-based PMW and PR observations to improve IR precipitation

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retrieval across the full-disk view, aiming to achieve precipitation retrieval over a broader spatial extent [63]. Additionally, we will focus on extreme precipitation events by incorporating more microphysical mechanisms, while exploring image generation methods to refine the smoothing of retrieval results and address spatial variability issues.

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